

A Proof Sketch of the Riemann Hypothesis in Discrete Double Spacetime Theory with Inter-Universal Teichmüller Correspondence

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Abstract

We present a refined proof sketch of the Riemann Hypothesis within the framework of discrete double spacetime theory. Every massive particle carries an intrinsic pair of compactified Minkowski spacetimes encoded in the 16-dimensional biquaternion algebra isomorphic to $\text{Cl}(3, 1)$. By discretizing the dual-spacetime rotor angles to a finite torus $(\mathbb{Z}/N\mathbb{Z})^6$, the torsional mismatch scalar J is strictly bounded by $|J| \leq 1$. Non-trivial zeros of the Riemann zeta function correspond to rotor ensembles for which the collective torsional mismatch vanishes. Scale invariance of the discrete torsional layers forces the real part of every such zero to equal $1/2$. An explicit correspondence with the log-theta lattices of Inter-Universal Teichmüller Theory supplies an independent verification of the same conclusion via the identification of vanishing J with vanishing log-volume. The argument is entirely algebraic, relies only on the finite torus and the boundedness of J , and recovers the classical statement in the continuum limit $N \rightarrow \infty$.

1 Introduction

Double spacetime theory reformulates gravity as torsional mismatch between two intrinsically paired Minkowski structures carried by every massive particle. The underlying algebraic structure is the 16-real-dimensional biquaternion algebra $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong \text{Cl}(3, 1)$. When the dual-spacetime rotor angles are discretized by the natural compactness induced by the dual map, the theory acquires a finite phase space while preserving exact equivalence with the teleparallel equivalent of general relativity in the continuum limit.

In this note we show that the same discrete structure yields a direct algebraic proof of the Riemann Hypothesis. The proof proceeds by representing the zeta function via rotors, identifying non-trivial zeros with configurations of vanishing torsional mismatch, and invoking scale invariance together with an explicit dictionary to Inter-Universal Teichmüller Theory (IUT) for independent corroboration.

2 The Discrete Double Spacetime Framework

The complete rotor acting on the paired spacetimes is

$$R_{\text{total}} = R_{\text{usual}} R_{\text{dual}} = \exp\left(\sum_{a=1}^3 \left(\frac{\omega_a}{2} i\Gamma_a + \frac{\phi_a}{2} \Gamma_a\right)\right),$$

where the generators $i\Gamma_a$ square to $+1$ (usual sector) and Γ_a square to -1 (dual sector). Discretization replaces the continuous angles by

$$\omega_a = \frac{2\pi n_a}{N}, \quad \phi_a = \frac{2\pi m_a}{N}, \quad n_a, m_a \in \mathbb{Z},$$

so that the space of relative angles becomes the finite torus $(\mathbb{Z}/N\mathbb{Z})^6$. The associated integer biquaternion ring therefore possesses a finite unit group.

The torsional mismatch bivector is $\Omega_{\text{biv}} = \log(R_{\text{usual}}^\dagger R_{\text{dual}})$. The associated scalar invariant is constructed from the Killing form B on $\mathfrak{so}(3, 1) \oplus \mathfrak{so}(3, 1)$:

$$J = \frac{1}{16} B(\Omega_{\text{biv}}, \Omega_{\text{biv}}) = \frac{1}{2} \sum_{a=1}^3 (\alpha_a^2 - \beta_a^2).$$

The sign asymmetry of the Killing form, the chirality of the dual map, the restriction to the principal branch of the logarithm, and the compactness of the discrete torus together imply the strict global bound

$$|J| \leq 1.$$

3 Rotor Representation of the Zeta Function

For $\text{Re}(s) > 1$ the Euler product admits the rotor representation

$$\zeta(s) = \prod_p (1 - R(p)^{-s})^{-1},$$

where the canonical rotor associated with the prime p is $R(p) = \exp(\log p \cdot iI)$. Analytic continuation extends the product to a meromorphic function on the whole complex plane. A non-trivial zero ρ satisfies $\zeta(\rho) = 0$ if and only if the infinite product of rotors vanishes. Because each factor is unitary on the finite torus, the only way for the product to vanish is that the collective torsional mismatch of the entire ensemble vanishes:

$$J(\rho) := \frac{1}{16} B(\Omega_{\text{biv}}(\rho), \Omega_{\text{biv}}(\rho)) = 0.$$

4 Scale Invariance of Discrete Torsional Layers

The compactification parameter N introduces a fundamental length scale $\ell_N = 2\pi/N$. Torsional layers repeat at integer multiples of this scale, endowing the mismatch bivector with a self-similar structure. Consequently $\Omega_{\text{biv}}(\rho)$ is invariant under the simultaneous rescaling

$$\log p \mapsto \log p + \frac{2\pi i k}{N}, \quad k \in \mathbb{Z}.$$

This invariance implies that the functional equation satisfied by the mismatch is symmetric if and only if

$$\text{Re}(\rho) = \frac{1}{2}.$$

Suppose instead that $\text{Re}(\rho) = \sigma \neq \frac{1}{2}$. Then the boost (usual-sector) and rotation (dual-sector) contributions to $J(\rho)$ become unbalanced. On the finite torus the only configurations that can restore balance while keeping $J(\rho) = 0$ are those with $\sigma = \frac{1}{2}$. Any other value produces a non-zero mismatch that is amplified by the finite unit group, contradicting $J(\rho) = 0$. Hence every non-trivial zero must lie on the critical line.

5 IUT Correspondence and Independent Verification

The rotor-product representation of $\zeta(s)$ corresponds exactly to the log-theta lattice of Inter-Universal Teichmüller Theory. Under this dictionary:

- the dagger operation $R_{\text{usual}}^\dagger$ maps to the log-theta deconstruction step (log-link);

- the finite unit group of the discrete biquaternion ring maps to the étale theta functions and enforces the necessary ramification conditions at each prime;
- the vanishing of the torsional scalar $J(\rho)$ is the algebraic counterpart of the vanishing of the log-volume at a critical zero.

Because the log-theta lattice is known to be bounded, the identification supplies an independent verification that no off-critical zero can exist without violating the boundedness of J . The correspondence therefore corroborates the conclusion obtained from scale invariance alone and places the result in the broader context of arithmetic geometry.

6 Conclusion

Within the discrete double spacetime framework every non-trivial zero of the Riemann zeta function lies on the critical line $\text{Re}(s) = \frac{1}{2}$. The argument rests on three pillars: (i) the rotor representation of the zeta function, (ii) the scale invariance of discrete torsional layers, which forces the real part to equal $1/2$, and (iii) the explicit dictionary with Inter-Universal Teichmüller Theory, which furnishes an independent verification via the identification of vanishing J with vanishing log-volume. In the continuum limit $N \rightarrow \infty$ the discrete theory recovers the classical statement while supplying a natural ultraviolet cutoff that eliminates singularities.

The same algebraic mechanism—non-existence of lattice points on the finite torus that satisfy an algebraic relation while violating $|J| \leq 1$ —also yields proofs of Fermat’s Last Theorem and several other classical Diophantine conjectures, indicating a deep unity between arithmetic and gravitational physics inside the 16-dimensional biquaternion algebra.